Persistent spectral localization and cylindrical geometry across the Kepler LEGACY sample

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Stage 11 evidence-aware working draft

Abstract

We define a star-grouped validation of localized power-spectrum persistence and classical seismic estimators for the 66-star Kepler LEGACY cohort. The current scientific gate is population_pending. The archived development evidence contains 1 stars with estimates and 10 verified mission products. Population generalization is not established until the signed confirmation release is executed once. This draft states the construction, planned comparisons and current limitations without substituting missing measurements with simulations.

1 Population and observation operator

Use the existing grouped split: 39 development, 13 internal validation, 13 final holdout stars and the previously studied descriptive benchmark Rudy. Monthly PDCSAP light curves follow PP-01. A common physical frequency grid is rebinned without extrapolation; monthly spectra are averaged with retained-exposure weights. Published global references are traceable to Silva Aguirre et al. (2017); mode/ratio references use the corrected Lund et al. (2017) erratum edition. The observational references are documented in [3, 2, 1].

2 Construction and controls

For a background-normalized scalar spectrum, path-superlevel H_0 persistence equals elder-rule peak prominence on the shared domain. We therefore report the independent prominence check as a mandatory control. A persistence-weighted peak centroid estimates $\nu_{\max}$ within a data-selected envelope. Comb trials around the classical spacing retain half, nominal and double alternatives. Cylinder scores compare periodic ridge geometry and finite H_0 localization but do not establish independent spectral information or blind degree identification.

3 Statistical analysis

The primary endpoints are errors in $\nu_{\max}$ and $\Delta\nu$. Both methods use the same successful stars; method-specific missingness, failures and catastrophic errors are displayed beside bias, MAE and RMSE. Star bootstrap draws are paired. Practical equivalence uses paired TOST on absolute fractional-loss differences, with registered margins 0.01 and 0.005 and Holm correction across both endpoints. Small-cell operating domains remain descriptive. Monthly intervals refer to the mean-spectrum estimator; reference-center coverage and reference-interval overlap are distinct quantities. A positive-only selected cohort cannot estimate specificity, false-alarm rates or probability calibration.

4 Current evidence and confirmation state

Role	Stars	Measured	Missing/failed
development	39	0	39
internal_validation	13	0	13
descriptive_benchmark	1	1	0

Machine-readable catalogs and reason-coded failure/operating-domain tables are generated separately for each role. The one-time holdout archive will supply final performance estimates; no population superiority claim is currently supported.

5 Ridge ratios and counterexamples

The degree/order-conditioned mode-localization benchmark compares r_{02} and related ratios to corrected individual-frequency references. It is explicitly reference-conditioned. Unanchored ridges, alias ambiguity, sparse orders, background instability and missing observations are failure cases, not suppressed examples. A ratio of median frequencies need not equal the median ratio. No covariance-based ratio interval is claimed without covariance data.

6 Domain of validity and reproducibility

This benchmark concerns selected main-sequence and mildly evolved solar-like LEGACY oscillators. Extrapolation to undetected oscillators, red giants or TESS requires separate validation. Delay topology remains dropped after Stage 7; cylindrical inference remains geometry-only after Stage 8. Earlier independent proof and human reproduction reviews remain pending. Code, environment, inputs, exclusions, thresholds and statistical rules must be signed and frozen before confirmation. Public identifiers are visible; local guards are procedural access controls, not cryptographic blinding. See the Stage 11 protocol, reproduction guide and task traceability for commands and exact archived evidence.

References

[1] M. N. Lund et al. Erratum: Standing on the shoulders of dwarfs. *The Astrophysical Journal*, 850:110, 2017.

[2] Mikkel N. Lund, Víctor Silva Aguirre, Guy R. Davies, et al. Standing on the shoulders of dwarfs: The kepler asteroseismic legacy sample. i. oscillation mode parameters. *The Astrophysical Journal*, 835(2):172, 2017.

[3] Víctor Silva Aguirre, Mikkel N. Lund, H. M. Antia, et al. Standing on the shoulders of dwarfs: The kepler asteroseismic legacy sample. ii. radii, masses, and ages. *The Astrophysical Journal*, 835(2):173, 2017.